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Four reasons to live well with the incomputability of complexity

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Problem: The incomputability of $C(s)$ seems to ruin the concept. For superficial observers, Kolmogorov complexity may have a whiff of mysticism. Compressing programs would be more tangible.

Question: Is complexity just a fancy name to mean efficient compression?

Answer: No! Complexity is a fundamental mathematical notion, independently from its approximation through compressors.

Algorithmic Information Theory is an essential theoretical framework, regardless of its implementations.
Let's mention four reasons for this.

1. $C(s)$ as it stands is essential to establish important **theoretical results**. We will see, for instance, that it can be used to prove Gödel's first incompleteness theorem in no more than a couple of lines! It also offers a conceptual framework with far-reaching consequences, in mathematics, in information science, in artificial intelligence, in epistemology, in philosophy and in Cognitive Science.
2. Restricted definitions of $C(s)$ are computable. It is obviously the case when we consider **resource-bounded** complexity, where $C(s)$ is the length of the shortest description obtained *so far*. It is also true for **approximations** of $C(s)$ based on compressors or on algorithmic probability. The ideal (non computable) definition of complexity provides a lower bound for practical complexity computations.
3. Results from complexity theory can be applied to specific "machines" of special interest. This is true, for instance, of computational (and computable) models of **human performance**.
4. Few people have noticed that other prominent theories such as **probability theory** can be regarded as incomputable as well. For instance, probability theory refers to sets (the set of people, the set of windows) that are not guaranteed to be computable (i.e. if there is no program to decide on

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